Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Summary of the problem found in the DNS and ICMP traffic log

The UDP protocol reveals that: The affected network traffic involved DNS requests sent over UDP. The browser attempted to contact the DNS server to resolve the domain name www.yummyrecipesforme.com into an IP address.

ICMP error message: “udp port 53 unreachable.”

Port usage: Port 53 is used by the DNS service, which translates domain names into IP addresses.

Most likely issue: The DNS server was not accepting requests on UDP port 53. This suggests the DNS service was down, misconfigured, or blocked.

Summary of tcpdump analysis: Multiple UDP DNS queries from 192.51.100.15 to 203.0.113.2 were observed. Each request resulted in an ICMP error indicating port 53 was unreachable, confirming a persistent DNS service issue.

Part 2: Analysis of the incident and likely cause

Time incident occurred: 13:24:32.192571

How the IT team became aware: Customers reported being unable to access the website and receiving a “destination port unreachable” error. The analyst reproduced the issue.

Investigation actions: The analyst used tcpdump to capture network traffic and identified repeated DNS queries over UDP followed by ICMP error responses.

Key findings: DNS queries failed due to “udp port 53 unreachable.” The DNS server did not respond properly, indicating service unavailability.

Likely cause: DNS service outage, misconfiguration, or firewall blocking UDP port 53.

Current status: The issue has been identified and escalated to the security engineering team.

Next steps: Verify DNS service status, check firewall rules, review server logs, and restore DNS functionality.

Final Conclusion

The incident affected the DNS protocol over UDP port 53. DNS queries failed due to ICMP errors indicating the port was unreachable, preventing domain resolution and access to the website.